

```
knitr::opts_chunk$set(fig.width = 4, fig.height = 3)
```

```
xbar <- 1.697
# sample mean
n <- 100
# sample size
alpha <- 0.01
# 99% = 1 - 0.01
se <- 0.088 / sqrt(n)
# standard error
z.star <- qnorm(1 - alpha / 2) # critical value
ME <- se * z.star
# margin of error
c(xbar - ME, xbar + ME)
```

```
## [1] 1.674333 1.719667
```

```
# confidence interval (CI)
```